Q-BRIDGE: Verifiable Policy Queries over Joint Confidence Views

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***Abstract*—Analysts combine production logs with randomized experiments to choose a policy, a rule for assigning actions, that improves expected outcomes over a reference. Selection bias, finite samples and limited transfer between populations leave action outcomes uncertain. When candidate policies cannot be certified, it is unclear whether another policy has the requested guarantee or the evidence supports none. Our approach separates this evidence limit from candidate choice by treating all policies as queries over one shared set of plausible action-success probabilities. Q-BRIDGE materializes this joint confidence view and returns independently checkable bounds for policy evaluation, redesign and stopping. For disjoint groups with nonnegative weights, a three-step construction finds a policy whose worst-case gain matches an upper bound for the entire family. Certificates establish the match against the original constraints; repeated fixed-policy queries reuse constraint weights and exact arithmetic. The construction agrees with an exact linear program on all 2,160 distinct synthetic family requests. Among 107 undecided primary views, full-family search finds 86 above-threshold policies and excludes such guarantees in 21; exact reoptimization of the original three policies recovers none of the 86. Controlled comparisons show that reusing exact arithmetic results lowers group-average query cost. Coat data illustrate redesign and stopping under declared assumptions.**

***Index Terms*—query processing, uncertain data, materialized views, causal inference, verifiable computation**

I. Introduction

A recommendation analyst combines production logs and a smaller randomized experiment to decide which items to recommend. A *policy* assigns action probabilities; its gain compares expected success with a reference.

Production exposure may depend on unrecorded preferences, while experimental information may transfer only for group averages. The data therefore permit multiple success probabilities. If a candidate lacks a guarantee, should the analyst try another policy, or can no policy guarantee the requested gain?

Our insight: separate candidate choice from the evidence limit. Q-BRIDGE[1] stores individual intervals and group constraints as one *confidence view*: the allowed mean vectors. A *robust guarantee* holds throughout this set. The same view supports candidate evaluation, optimal policy search and stopping when no guarantee exceeds a requested margin. Each bound carries numerical evidence, a *certificate*, independently checked against the view.

Building on causal systems [1], [2] and uncertainty-aware queries [3], we materialize evidence once, answer changing policy queries without rescans, and reuse independently checkable computation.

One example throughout. Figure 1 uses four unknown success probabilities $m_A, \ldots, m_D$ and reference probability $1/4$ per action. Observations give intervals; randomization gives $(m_A + m_C)/2 = .65$ and $(m_B + m_D)/2 = .70$. The reference's success probability is $(.65 + .70)/2 = .675$. Since $m_A \leq .6$, the first group forces $m_C \geq .7$. Always choosing C guarantees $.7 - .675 = .025 = 2.5$ percentage points (pp). The allowed vector $(.6, .7, .7, .7)$ limits every action to .7, ruling out a larger guarantee. Matching bounds proves optimality.

Figure 1(b) tests a margin $\delta = .001 = 0.1$ pp. Section IV-D works through these three policies to explain when saved evidence suffices and when a new solve is needed.

One view for changing policies (II–III). Stored constraints combine source assumptions and simultaneous intervals. When they cover the true means, worst-case gains bound actual gains for all policies, including adaptively chosen ones.

Search beyond the candidates (IV). For disjoint groups with nonnegative weights, three steps cap action means, allocate below that cap, and recover an attaining policy. We prove that the smallest feasible cap preserves optimality and recover certificates using only original constraints. A general exact LP supplies the same contract for overlaps.

Policy benefit and execution cost (V–VI). All 86 recovered guarantees require different policies. Retaining exact terms lowers group-average cost with answers and solver paths fixed. Coat connects certificates to redesign and catalogue queries.

Table I summarizes the interface.

II. Policy Queries and Their Guarantees

Two questions are distinct: does the view contain the true means, and does a proposed bound hold over that view? The first is statistical; the second is computational.

[1]Code and saved certificate examples: https://github.com/yzd2002/Q-BRIDGE-ICDE2027.

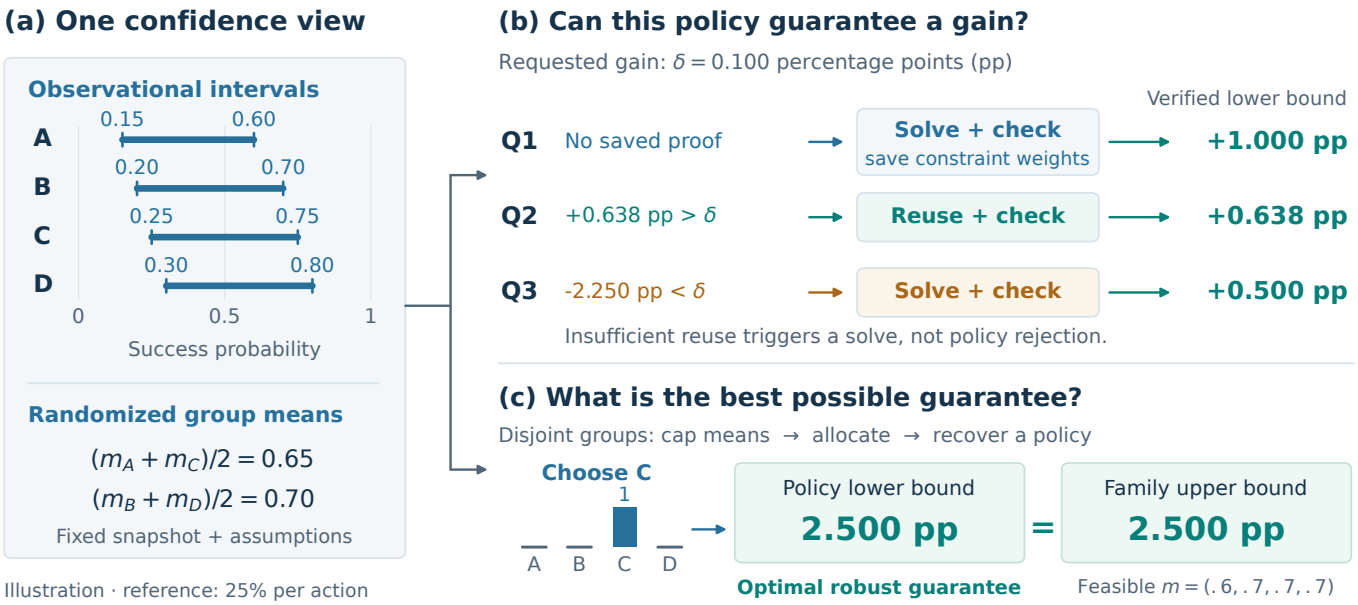


Fig. 1. One view, two query paths. (a) Store the allowed success probabilities. (b) Check a proposed policy: Q2 reuses a certificate; inconclusive reuse for Q3 triggers a solve. (c) Search all policies: matching bounds certify the optimal guarantee of 2.5 pp, not the actual effect. Teaching inputs are constructed and rounded; exported proofs use exact represented inputs.

TABLE I
THREE QUESTIONS OVER ONE FIXED VIEW. GUARANTEES CONCERN EVERY ALLOWED MEAN VECTOR; THEIR FAILURE DOES NOT ESTABLISH A NONPOSITIVE ACTUAL EFFECT.

Question	Answer and independently checked evidence
Does this policy guarantee gain $> \delta$?	True: a lower bound above δ. False: an allowed counterexample. Unknown: neither established. Reuse or solve for this policy; overlapping groups allowed.
What is the largest worst-case gain?	Capacity $\mathcal{R}$ and an attaining policy: its lower bound matches an upper bound for all policies. Our direct construction handles disjoint nonnegative groups; a general exact LP also handles overlaps.
Can every such guarantee be ruled out?	Yes, if one allowed mean vector limits even the best policy's gain to δ. No full optimization is needed when this witness suffices.

A. The query and the stored object

Let $X \in \{1, \ldots, K\}$ be context (e.g., user group), $A \in \{1, \ldots, L\}$ action, $Y \in \{0, 1\}$ success, and $d = KL$. The potential outcome $Y(a)$ is success under action a. Observational source O defines the target population; randomized source R may describe another population. Their means are $m^O_{xa} = \mathbb{E}_O[Y(a) \mid X = x]$ and $m^R_{xa} = \mathbb{E}_R[Y(a) \mid X = x]$.

Policy π_{xa} is the probability of taking action a in context x; π_0 is the reference and w_x the target context proportion. The expected gain is

$$\Delta(c) = \sum_{x,a} w_x(\pi_{xa} - \pi_{0,xa}) m^O_{xa} = c^\top m^O. \tag{1}$$

Thus $c_{xa} = w_x(\pi_{xa} - \pi_{0,xa})$ records where action probability increases or decreases. Each policy row and w sums to one and is caller-specified; uncertainty in estimated weights is outside this contract.

Flatten the context–action pairs into $i = 1, \ldots, d$. The view stores individual bounds ℓ, u and J weighted-group bounds a, b with weight matrix T:

$$\widehat{\mathcal{C}} = \{m : \ell \le m \le u,\ a \le Tm \le b\}. \tag{2}$$

Each m is a possible target-mean vector. Figure 1's two rows average A/C and B/D, coupling their means rather than treating intervals independently.

The *worst-case gain* is $V(c) = \min_{m \in \widehat{\mathcal{C}}} c^\top m$. A query tests $V(c) > \delta$, usually $\delta = .001 = 0.1$ pp. Section III covers the true m^O with probability at least $1 - \alpha$; on this event, $V(c)$ bounds actual gain. Section IV verifies bounds on $V(c)$ against stored constraints, not the source assumptions.

Data and declarations. Events (x, a, y) are iid within each source; panels need user-level treatment (Section VI-D). Counts are N^s_{xay}, with $N^s_{xa} = \sum_y N^s_{xay}$ and $S^s_{xa} = N^s_{xa1}$. Randomization requires $A \perp Y(a) \mid X$ and known $q_{xa} = P_R(A = a \mid X = x) > 0$ on moment supports. A union bound allows dependence across sources and rows.

Prespecify contexts, target, source assumptions, rows, sensitivities and confidence allocation, optionally from independent design data. Objectives and thresholds may adapt; new snapshots or confidence constructions need another guarantee.

B. Bounding observational selection bias

Recipients' success rate $p_{xa} = P_O(Y = 1 \mid x, a)$ can differ from the target rate m^O_{xa}. Reweighting corrects this selection. For unobserved factors U and exposure probabilities e_O, consistency, ignorability given (X, U) and latent overlap give

$$r^*(x,a,y) = \mathbb{E}_O\left[\left.\frac{e_O(a \mid x)}{e_O(a \mid x, U)}\right| x, a, y\right],$$
$$m^O_{xa} = \mathbb{E}_O[r^* Y \mid x, a]. \tag{3}$$

The theoretical weight r^* averages exposure corrections within outcome groups. Declaring $1/\Lambda \le r^* \le \Lambda$, $\Lambda \ge 1$, permits more bias and wider bounds as Λ grows. It bounds an outcome-conditional average, not propensity odds or every latent weight. No propensity model is supplied; data do not justify Λ [4].

For binary outcomes, these weight bounds translate directly into a permitted interval for the target mean [5, Eq. (9)]:

$$L_\Lambda(p) = \max\{p/\Lambda, 1 - \Lambda(1-p)\},$$
$$U_\Lambda(p) = \min\{\Lambda p, 1 - (1-p)/\Lambda\}. \tag{4}$$

Writing $r_y = r^*(x,a,y)$, $m = m^O_{xa}$ and $p = p_{xa}$, normalization $\mathbb{E}[r^* \mid x, a] = 1$ gives $r_1 = m/p$ and $r_0 = (1-m)/(1-p)$ for $0 < p < 1$. Bounding both yields (4), exact for this relaxation [6].

Consistency provides additional information. Outcomes under a are known for those who received it; the remaining potential outcomes may be zero or one. Thus

$$P_O(A = a, Y = 1 \mid x) \le m^O_{xa} \le 1 - P_O(A = a, Y = 0 \mid x). \tag{5}$$

These probabilities use all people in context x as the denominator, not just those receiving a. Both fusion and OBS include these joint-count constraints. With exposure .8 and $p = .5$, they give $[.4, .6]$, versus $[.25, .75]$ from $\Lambda = 2$ alone, without fitting propensities. This changes a substantive baseline comparison (Section VI-C).

C. Which randomized information transfers?

An analyst may declare a jacket-type average stable across sources while individual item means change. Row j selects context x_j, actions S_j and nonnegative weights ν_{ja} summing to one. The normalized matrix H has $H_{j,(x_j,a)} = \nu_{ja}$ on this group and zero elsewhere. It is the exact row-normalization of stored T, fixed before inference (III-C). The declaration is

$$\tau_j = (Hm^R)_j, \qquad |(Hm^O)_j - \tau_j| \le \rho_j. \tag{6}$$

Here ρ_j permits source differences in a group mean; zero requires equality. Thus Λ controls selection bias, ρ transfer differences, and α sampling error. Two positive-weight action means can shift by $(h, -\nu_1 h/\nu_2)$ while preserving their average: group agreement does not establish action-level agreement.

With exact population group means and zero tolerance, $c = H^\top v$ for a row-weight vector v implies $c^\top m = v^\top \tau$: the query is a weighted sum of known group means. Finite samples replace τ by intervals. Components outside this row span can remain variable within the box. Full cell transport ($H = I_d$, $\rho = 0$) is stronger; Coat includes its direct-evaluation control.

III. Building One Confidence View

For independent binary events, each source's $2d$ counters determine score moments and, with fixed declarations, the bounds in (2). Aggregation costs $O(n_O + n_R)$ work and $O(d)$ storage; new policies need no rescan. User panels also need group-score variances (Section VI-D).

A. Observational intervals and an explicit budget

Here a confidence *budget* is permitted failure probability, not sample size. Split $\alpha = \alpha_O + \alpha_R$ between sources. Of α_O, half covers d two-sided Clopper–Pearson (CP) intervals [7] $[p^-_{xa}, p^+_{xa}]$, each at $\alpha_O/(2d)$. The other half covers $2d$ one-sided joint-event bounds in (5), each at $\alpha_O/(4d)$, with denominator $N^O_x = \sum_{a,y} N^O_{xay}$. Denote their lower endpoints g^1_{xa}, g^0_{xa}. Then

$$\ell_{xa} = \max\{L_\Lambda(p^-_{xa}), g^1_{xa}\},$$
$$u_{xa} = \min\{U_\Lambda(p^+_{xa}), 1 - g^0_{xa}\}. \tag{7}$$

Monotonicity makes propagation conservative. Conditional binomial coverage permits random stratum counts. Empty action cells contribute $[0, 1]$; empty contexts add no joint restriction. The union bound permits interval dependence.

B. Two prespecified randomized constructions

Both routes below produce an interval $[\tau^-_j, \tau^+_j]$ for the same experimental group mean. Transfer expands it by ρ_j on either side for the target mean; III-C converts this restriction to stored row T_j. Allocate α_R to one route, chosen before outcomes: average individual-mean intervals, or estimate the group mean directly. The synthetic primary comparison uses the former; Coat reports both.

Projected cell intervals. Construct simultaneous binomial intervals $[r^-_i, r^+_i]$ for every randomized cell appearing in H, splitting α_R across those cells. Nonnegativity of H gives normalized group intervals $[Hr^-, Hr^+]$. Only these *group* intervals enter (6); individual experimental means are not equated to target means. Selecting the tighter route after seeing data is not covered by either individual budget.

Group scores. Conditional on an event falling in (x_j, S_j), define $q_j(a) = q_{x_j a}/\sum_{b \in S_j} q_{x_j b}$ and $Z_j = \nu_{jA} Y / q_j(A)$. This reweights experimental assignment to the desired group weights: $\mathbb{E}[Z_j] = \tau_j$ and $0 \le Z_j \le B_j = \max_a \nu_{ja}/q_j(a)$. Let n_j count these events and $\bar{Z}_j$ be their average score. Counts, sums and squared sums follow from N^R_{xay}, since $Y^2 = Y$. If every multiplier $\nu_{ja}/q_j(a)$ equals one, $Z_j = Y$ and we use a binomial interval. Otherwise, for $n_j \ge 2$, the empirical Bernstein interval [8] has half-width

$$h_j = \sqrt{\frac{2\widehat{\mathrm{Var}}(Z_j)\log(4/\beta_j)}{n_j}} + \frac{7B_j \log(4/\beta_j)}{3(n_j - 1)}. \tag{8}$$

Here $\beta_j = \alpha_R/J$, and the sample variance uses $n_j - 1$. The two terms account for score variation and its bounded range. The interval is $[\bar{Z}_j - h_j, \bar{Z}_j + h_j] \cap [0,1]$: only the expectation interval is clipped, not the scores. Insufficient observations use the full range. The union bound covers overlapping groups sharing events.

C. Preserving the statistical meaning in stored coefficients

The statistical average and its stored row must describe the same problem. The checker treats each double-precision (binary64) coefficient as an exact rational. Five stored values of .2 sum to $1 + 2^{-54}$; assuming unit mass would break the coverage link.

For nonzero T_j, its exact mass $\kappa_j = \sum_i T_{ji}$ defines $H_{ji} = T_{ji}/\kappa_j$ and $T_j m = \kappa_j H_j m$; zero rows impose deterministic zero constraints. Assignment probabilities are either exact or explicitly normalized nonnegative design weights. Their match to the data-generating assignment law is assumed, not established by normalization.

For a covered normalized interval and tolerance ρ_j, multiply by κ_j, rounding the lower endpoint down and the upper endpoint up:

$$\begin{aligned} a_j &= \operatorname{down}\left[\kappa_j(\tau_j^- - \rho_j)\right], \\ b_j &= \operatorname{up}\left[\kappa_j(\tau_j^+ + \rho_j)\right]. \end{aligned} \tag{9}$$

Projected intervals use exact $T_j r^-$ and $T_j r^+$ expanded by $\kappa_j \rho_j$. Observational endpoints, complements, score moments and normalization use rational arithmetic with one outward endpoint rounding; exact empty intervals are refused first. CP quantiles and all floating-point Bernstein arithmetic remain trusted, including logarithms and square roots. Neither these routines nor scientific declarations are formally verified.

D. Why one view supports many queries

Set $\alpha_O + \alpha_R = \alpha$ (equal halves for fusion here). The endpoints cover the true means together. On this event, minimization over the view bounds every policy, without a new confidence budget per query.

Theorem 1 (One event, all objectives): Under the source and sampling conditions in Section II, $P(m^O \in \widehat{\mathcal{C}}) \geq 1 - \alpha$. On that event, every linear objective, including one chosen adaptively from previous answers, satisfies $V(c) \leq c^\top m^O$. Any verified lower approximation to $V(c)$ shares this inequality. A method that cannot establish a usable view or bound makes no positive assertion.

Proof. Conditional on independent design information, interval coverage averaged over stratum counts and monotone mapping cover the box at $1 - \alpha_O$. Prespecified randomized intervals cover Hm^R at $1 - \alpha_R$; transport and outward mapping give target moments. A union bound covers m^O at $1 - \alpha$. On that event, minimizing over the view bounds every objective simultaneously, including adaptive choices. Weak duality extends this to checked lower bounds. □

Reuse retains the statistical design [9], [10]. View identity binds count digests, confidence allocation, assignment design, radius, tolerances and matrix; protocols retain provenance.

IV. Query Execution and Verification

A *producer* proposes evidence; an independent *receiver* checks it against the caller's view $\mathcal{C} = \widehat{\mathcal{C}}$. We check one policy (A), optimize the family (B–C), then reuse evidence across policies (D–E).

A. Checking a proposed policy

A usable view first needs an exactly checked feasible point, proposed by a solver or slack maximization. Otherwise the status is `uncertain`: no certificate or decision. This gate prevents empty-view certification but neither proves emptiness nor reliably accepts zero-volume sets.

Together with a lower certificate, a checked feasible point brackets the optimum [11, Sec. 5.5.1]:

$$L(c) \leq V(c) \leq U(c) = c^\top \widetilde{m}. \tag{10}$$

Return true for $L > \delta$, false for $U \leq \delta$, and unknown otherwise. False rejects a guarantee, not the actual effect; a nonpositive lower bound decides nothing. Artifact status `unresolved` records no positive lower certificate and can accompany false or unknown. Checked points remain feasible across objectives: retain them after global checks, round later dot products upward, and bind exported evidence to the view and objective.

How a lower certificate works. Give the lower and upper group constraints nonnegative weights η_L, η_U. For any feasible m, both $\eta_L^\top(a - Tm)$ and $\eta_U^\top(Tm - b)$ are nonpositive. Adding them to the gain can only lower it. With $z = c + T^\top(\eta_U - \eta_L)$, minimizing this adjusted expression over the individual intervals gives

$$D(c;\eta) = a^\top \eta_L - b^\top \eta_U + \sum_i \min\{z_i \ell_i, z_i u_i\} \leq V(c). \tag{11}$$

Each term chooses the cheaper endpoint: ℓ_i when $z_i \geq 0$, u_i otherwise. Thus a checker needs the weights, current objective and view, not a replay of the optimization. Any nonnegative weights give a valid lower bound $L = D(c;\eta)$; better weights tighten it [12].

The standard-library checker computes exactly on represented coefficients and rounds the bound downward. Proofs bind objective, threshold and view digest; the caller supplies the trusted view. Nonfinite or failed proposals cannot authorize success. Independent optimization certificates are established [13].

B. Certifying a whole policy family

Testing one policy can miss a better one. The family Π contains all randomized policies on the available actions; its *capacity* is the largest worst-case gain. Fix a nonempty compact convex view $\mathcal{C} \subseteq [0,1]^{KL}$, exact probability weights $w \geq 0$, $\sum_x w_x = 1$, and reference $\pi_0 \in \Pi$. Stars will denote returned optimizers. Define

$$\mathcal{R}(\mathcal{C}) = \max_{\pi \in \Pi} \min_{m \in \mathcal{C}} \sum_{x,a} w_x(\pi_{xa} - \pi_{0,xa}) m_{xa}, \tag{12}$$

$$F(m) = \sum_x w_x \left[\max_a m_{xa} - \sum_a \pi_{0,xa} m_{xa}\right]. \tag{13}$$

The inner minimum gives one policy's worst case; the outer maximum chooses the best policy. The reference ensures $\mathcal{R} \geq 0$. At one allowed m, $F(m)$ chooses the best action per context. It bounds $\mathcal{R}$ because each policy's minimum cannot exceed its payoff at this point. Thus $U(c)$ bounds one policy's guarantee; $F(m)$ bounds every policy's guarantee. A checked $F(m) \leq \delta$ excludes all above-threshold guarantees, not actual effects.

Compact convex domains and a continuous bilinear payoff permit minimax exchange [14], giving $\mathcal{R}(\mathcal{C}) = \min_{m \in \mathcal{C}} F(m)$. We therefore seek an allowed mean vector giving the best action as little advantage as possible, and a policy whose lower certificate reaches this upper bound.

C. Exact capacity for partitioned moments

For one context, minimize $t - p^\top m$: the largest action mean minus reference payoff. Here t caps every m_i and $p = \pi_{0,x}$ is the reference distribution. Step 1 finds the smallest feasible cap; Step 2 maximizes reference payoff below it; Step 3 recovers an attaining policy and *original*-constraint proof. Minimax and continuous allocation are classical [15], [16]; the key is to prove that this cap suffices and recover the certificate.

Theorem 2 (An attaining policy with an original-view certificate): Let $\mathcal{C}$ be a nonempty rational box-and-moment view with $0 \leq \ell \leq u \leq 1$. Suppose every nonzero moment row is nonnegative, belongs to one context, and has positive support disjoint from every other row. For any rational probability weights w and reference policy π_0, including zero entries, the three-step construction returns $m^* \in \mathcal{C}$, $\pi^* \in \Pi$ and nonnegative original-row multipliers η_L, η_U such that, with $c^*_{xa} = w_x(\pi^*_{xa} - \pi_{0,xa})$,

$$D(c^*; \eta) = c^{*\top} m^* = F(m^*) = \mathcal{R}(\mathcal{C}).$$

The dense implementation uses $O(dJ + d^2)$ exact arithmetic/comparison operations; the independent receiver uses $O(dJ + d)$. These are unit-cost rational-operation counts, excluding coefficient-dependent bit costs, encoding and hashing.

The certificate matches an achievable guarantee to the family upper bound: $D(c^*; \eta) \leq \min_{m \in \mathcal{C}} c^{*\top} m \leq \mathcal{R} \leq F(m^*)$. Equal checked endpoints prove optimality independently of the producer. Reject unsupported rows. Zero rows require $a_j \leq 0 \leq b_j$ and receive zero multipliers. Ties use index order. The following three steps establish the theorem.

Step 1: Find the smallest feasible cap.

Within the construction, $p = \pi_{0,x}$ is the reference distribution, distinct from observed success p_{xa}. A cap is feasible only if the capped upper endpoints can meet every lower group requirement. Define

$$\begin{aligned} t_\ell &= \max_i \ell_i, \qquad g_j(t) = \sum_i T_{ji} \min(u_i, t), \\ t_0 &= \min\{t \geq t_\ell : g_j(t) \geq a_j \quad \forall j\}, \\ h_i &= \min(u_i, t_0). \end{aligned}$$

The maximum group mass $g_j(t)$ is piecewise linear between rational upper-box breakpoints. The largest required root, or t_ℓ, gives t_0. Nonemptiness ensures $T_j \ell \leq b_j$. Disjoint groups independently choose mass in $[\max(a_j, T_j\ell), \min(b_j, T_j h)]$, so $\mathcal{C} \cap [\ell, h]$ is nonempty. Every point in this intersection has maximum t_0 by minimality; the entire box need not be feasible. Figure 1's group requirements give $t_0 = .7$ and $h = (.6, .7, .7, .7)$.

Minimum-cap sufficiency. A larger cap could raise the reference payoff, so feasibility alone is insufficient. For feasible m with $t = \max_i m_i$, clip to $v_i = \min(m_i, t_0)$, preserving lower boxes and upper moments. Restore violated lower moments within $[v, h]$ to $a_j \leq b_j$; disjoint supports prevent interference, giving feasible $v' \geq v$. Clipping lowers the maximum by $t - t_0$ and the reference average by at most that amount: $p^\top(m - v) \leq t - t_0$. Repairs only raise the average. Hence

$$\max_i v'_i - p^\top v' \leq t_0 - p^\top v \leq t - p^\top m.$$

Thus maximizing $p^\top m$ within this cap minimizes the original gap.

Step 2: Allocate means within the cap. With the maximum fixed at t_0, increasing the reference payoff reduces the gap. We allocate *means*, not policy probabilities. Raising m_i by ϵ costs group mass $T_{ji}\epsilon$ and gains $p_i\epsilon$. Starting at ℓ, fill each nonzero row in decreasing p_i/T_{ji} order on $T_{ji} > 0$, to total mass $\min(b_j, T_j h)$; uncovered coordinates take h_i. The added mass is $\min(b_j, T_j h) - T_j \ell$. Step 1 ensures the final total $\min(b_j, T_j h)$ meets a_j. Nonnegative weights justify filling even zero-ratio coordinates.

For Step 3, record an upper-row price λ_j: a charge per unit of group mass used to build the certificate. Use zero when $b_j \geq T_j h$, including equality; the cap already limits the total. Otherwise use the largest support ratio if $b_j = T_j \ell$, or the last ratio receiving positive mass, including exact boundary exhaustion. These rules cover fixed coordinates and zero probabilities. Subtracting the group charges leaves $\xi = p - T^\top \lambda$. Positive residuals fill to h_i; negative ones stay at ℓ_i:

$$\begin{aligned} \xi_i > 0 \Rightarrow m^*_i = h_i, \quad \xi_i < 0 \Rightarrow m^*_i = \ell_i, \\ \lambda_j(b_j - T_j m^*) = 0. \end{aligned}$$

Thus a positive row price requires a tight upper constraint. In the example, $m^* = (.6, .7, .7, .7)$ and $\lambda = (0, 0)$, giving upper bound $.7 - .675 = .025$.

Step 3: Recover the policy and original-row prices. Steps 1–2 give a family upper bound, not yet a robust policy. B ties for best at $m^* = (.6, .7, .7, .7)$, yet the allowed vector $(.6, .6, .7, .8)$ gives pure B a $-.075$ gain. Step 3 must match the family bound with an original-constraint policy lower bound. Two conditions make (11) equal this upper bound. The policy must sum to one and select only maximum coordinates of m^*. With lower prices $\eta_L = \eta$ and upper prices $\eta_U = \lambda$, its coefficient $z = \pi - p + T^\top(\lambda - \eta)$ must attain its original-box minimum at m^*: $z_i > 0$ requires $m^*_i = \ell_i$, and $z_i < 0$ requires $m^*_i = u_i$. Rows with positive prices must also be tight.

Start with a nonnegative vector that enforces these endpoint signs but may have missing probability mass:

$$\widetilde{\pi}_i = (\xi_i)_+ \mathbf{1}\{u_i \geq t_0\}, \qquad D_\pi = 1 - \sum_i \widetilde{\pi}_i \geq 0.$$

Here $0 \le \widetilde{\pi}_i \le p_i$, with positive mass only at maximum coordinates. The original-box coefficient $z_i^{\text{init}} = \widetilde{\pi}_i - \xi_i$ is positive only at ℓ_i and negative only at u_i: below the cap $h_i = u_i$; elsewhere policy mass cancels positive residuals.

If $t_0 = t_\ell$, add D_π to a coordinate with $\ell_i = t_0$ and set all lower prices η to zero; its box coefficient increases at this lower endpoint. Otherwise choose the first row whose required root is t_0, so it forces the cap. Since $T_j h = a_j$, its supported means must equal h_i, and zero-price priority gives $\lambda_j = 0$, even if its upper moment is also tight. Set

$$E = \{i : T_{ji} > 0,\ u_i \ge t_0\}, \qquad \sigma_E = \sum_{i \in E} T_{ji} > 0,$$

where positivity follows because otherwise this row would attain its bound below t_0. Set $\eta_j = D_\pi / \sigma_E$, with other lower prices zero. Add $\eta_j T_{ji}$ to $\widetilde{\pi}_i$ on E. Updated box coefficients are zero on E and $-p_i - \eta_j T_{ji} \le 0$ elsewhere in this row, where $m_i^* = h_i = u_i$; other coefficients are unchanged. The non-strict $u_i \ge t_0$ includes roots at breakpoints.

The completed π sums to one and selects maximum coordinates. With $z = \pi - p + T^\top(\lambda - \eta)$, box signs and tight priced rows make (11) exact for $c = \pi - p$ and $(\eta_L, \eta_U) = (\eta, \lambda)$:

$$\begin{aligned} D(\pi - p; (\eta, \lambda)) &= (\pi - p)^\top m^* \\ &= t_0 - p^\top m^*. \end{aligned}$$

Scale each context's objective and both row-price types by the same w_x. Concatenation yields original-program lower $= F(m^*) = \mathcal{R}(\mathcal{C})$, including zero-weight contexts. Such contexts still require a feasible point and a legal policy; only their objective and multipliers vanish.

The receiver checks original constraints, legal policy probabilities, multiplier signs and lower/objective/upper equality against the caller's request. It trusts neither the artificial cap nor the producer's internal state. □

In the example, $(0, .25, .25, .25)$ lacks .25. The A/C row forces the cap; within it, only C has $u_i \ge .7$. Thus $\sigma_E = .5$, $\eta_L = (.5, 0)$ and $\eta_U = 0$. Add .25 to C to obtain $\pi^* = (0, .25, .5, .25)$. The lower bound $.65(.5) - .6(.5) = .025$ equals the family upper bound $.7 - .675 = .025$. Pure C in Fig. 1 is another optimal policy. These decimal teaching inputs differ from represented inputs checked in measured runs.

Stopping without the full construction. A constant-action point makes every policy tie with the reference. This offers a cheap stopping certificate, including for overlapping groups.

Proposition 1 (Zero capacity): If all $w_x > 0$ and $\pi_{0,xa} > 0$, then $\mathcal{R}(\mathcal{C}) = 0$ exactly when $\mathcal{C}$ contains a vector constant across actions within each context. Without strict positivity, a constant point is sufficient but need not be necessary.

Proof. The nonnegative summands of F vanish, under strict positivity, exactly at constant rows. Compactness attains its minimum; the minimax identity gives the result. □

For nonnegative single-context moment rows, even overlapping ones, form

$$I_x = [\max_a \ell_{xa}, \min_a u_{xa}] \cap \bigcap_{j : x_j = x} [a_j/\kappa_j, b_j/\kappa_j].$$

Here $\kappa_j = \sum_i T_{ji} > 0$; zero rows require $a_j \le 0 \le b_j$. Choose $v_x \in I_x$, set every $m_{xa} = v_x$, and check the original constraints. Under strict positivity, an empty intersection proves $\mathcal{R} > 0$, but not $\mathcal{R} > \delta$ for positive δ.

Saved evidence can also bracket the family. The maximum of checked policy lower bounds and the reference's zero is $L_{\text{saved}} \le \mathcal{R}$. A nonempty set M_{saved} of checked points gives $U_{\text{saved}} = \min_{m \in M_{\text{saved}}} F(m) \ge \mathcal{R}$. Share only for identical $\mathcal{C}, w, \pi_0$. Then $L_{\text{saved}} > \delta$ certifies existence, $U_{\text{saved}} \le \delta$ excludes it, and otherwise the family remains unresolved. Equation (17) separates loose certificates from restricted policy choice.

D. Reusing certificates across policies

New objectives can reuse any nonnegative constraint weights: scaling changes tightness, not validity. A bounded first-in, first-out (FIFO) bank stores duals divided by their old objectives' binary64 ℓ_1 norms; the new norm supplies scale s. Zero old norm skips insertion; zero new norm gives $s = 0$. Floating scores order candidates; checking uses exact saved base–scale products. Retain group terms $k = a^\top \eta_L - b^\top \eta_U$ and $r = T^\top(\eta_U - \eta_L)$. For represented $s \ge 0$,

$$D(c; s\eta) = sk + \sum_i \min\{(c_i + sr_i)\ell_i, (c_i + sr_i)u_i\}. \quad (14)$$

Setup costs $O(\text{nnz}(T) + d + J)$ rational operations, later checks $O(d)$, with coefficient-dependent bit costs. Separately rounded multipliers can change last bits. A fresh receiver reconstructs (14); view/base-bound state is reconstructed or bypassed on validation failure. Dual and bounded-Lagrangian reuse are established [17], [18], [19]; persistent LP reoptimization supplies a control [20].

The example: solve once, then try reuse. Q1 changes the uniform policy to $(.15, .15, .35, .35)$, giving $c = (-.1, -.1, .1, .1)$. The two group means imply

$$\Delta_1 = .27 - .2(m_A + m_B) \ge .01.$$

The allowed vector $(.6, .7, .7, .7)$ attains it; interval-only and group-only bounds are $-.075$ and $-.13$. Q1's weights $\eta_L = (.2, .2), \eta_U = 0$ give $k = .27$, $r = (-.1, -.1, -.1, -.1)$.

Q2 has $c = (-.105, -.09, .105, .09)$ and norm .39, versus Q1's .4. Scaling Q1's unnormalized weights by $.39/.4 = .975$ is equivalent to the bank's normalization. Equation (14) gives .006375, passing .001 without finding the optimum .0105. Q3 has $c = (-.05, -.15, .15, .05)$ and scale 1. Reuse gives $-.0225$, inconclusive; a new solve gives .005. A weak lower bound triggers work, not rejection.

E. Execution choices and numerical safeguards

Validate the view, check ordered reusable bounds, and call the LP if needed. Candidates start with zero dual, last LP-solver (native) dual, initial feasible point and last native point; histories hold at most 16 of each enabled kind, including the last. Reuse hits preserve native last-result state. Native dual-simplex cutoffs [21] require exact checking; undecided requests resume without cutoff and may remain unknown.

Checked native duals update the bank; zero-dual answers are not dual reuse.

An objective cache serves the *same* c; the candidate bank serves *new* c. Retention caches exact k, r. Screening cheaply encloses a candidate's bound value with outward binary64 intervals. It defers a lower candidate if its interval's upper endpoint is $\leq \delta$, or a primal candidate if its lower endpoint is $> \delta$. Overflow or unbounded intervals require exact work. Screening cannot authorize answers.

True may bypass insufficient deferred lowers. Before false, replay deferred lowers to preserve the best exported lower; before solving, replay all remaining candidates in order. Evidence, native state and solver paths match the unscreened plan. Only zero dual before last-dual true, or initial primal before last-primal false, can be omitted permanently; complete service cost measures the benefit.

Structural controls use box endpoints, continuous knapsack for single rows [22], or sparse HiGHS for overlaps [23]; only the reconstructed global dual is certified.

From stored contrasts to intended policies. If intended rational c^{ideal} is stored as c, let $e = c^{\text{ideal}} - c$. Because $0 \leq m_i \leq 1$, the objective change lies between $\sum_i \min(e_i, 0)$ and $\sum_i \max(e_i, 0)$. Applying these corrections to the checked *minimum*-gain bracket gives

$$[L + \sum_i \min(e_i, 0),\ U + \sum_i \max(e_i, 0)]. \tag{15}$$

The additions and threshold test are exact; a positive stored bound need not certify an ideal zero-margin query. Fixed-query timings (V-D) use stored contrasts, family queries (V-B) exact probabilities, and Coat (VI-D) both with (15) as needed.

V. Evaluation Design

Three comparisons isolate policy choice, execution and information: vary policies on a fixed view (VI-A); vary reuse with four core plans' answers, evidence and solver paths fixed (VI-B); vary source construction and confidence allocation for a fixed policy (VI-C). Family analysis is retrospective; the source endpoint is registered. Coat tests redesign and stopping (VI-D). Exact-solver and new Coat timings use an AMD EPYC 7763, Python 3.12.3 and one CPU/native thread.

A. Fresh worlds and policy design

A world is a simulated data-generating distribution; each inference replicate is a fresh sample from it. The registered study draws 48 confounded worlds: $(K, L) \in \{(2, 4), (8, 8), (16, 16)\}$, uniform/skewed randomization and group drift $0/.02$, with four worlds per stratum and five inference samples per world. Twelve additional worlds test near-unit sensitivity. A latent binary factor affects exposure and rewards. Metadata $h_a \in [-.5, .5]$ are fixed without inference labels, but their designed positive reward association limits representativeness. Random groups allow individual shifts while controlling group drift; shifts are not optimized for the evaluation objective.

Inference uses $2{,}000d$ O events and $100d$ or $1{,}000d$ R events. Independent design data use $500d$ O and $100d$ R events. Against a uniform reference, the three policies are: $\text{softmax}(1.5h)$; $\text{softmax}(4\widehat{p}_x)$ with design-R $\widehat{p}_{xa} = (S_{xa} + .5)/(N_{xa} + 1)$; and the design-view choice with the largest checked positive lower bound among twelve tilts. The tilts use linear or alternating action scores and slopes $\{-2, -1, -.5, .5, 1, 2\}$; ties retain the first positive maximum, and otherwise the reference. Design outcomes never enter inference counts. We compare $\Lambda = 2$ with an oracle boundary computed from simulator truth and rounded upward. This diagnostic preserves counts and policy rules, but the design-selected policy can vary with radius; it does not isolate a radius effect or replace the registered comparison.

The primary endpoint compares projected fusion with certification by either full-α OBS or projected R, for the metadata policy, low R budget and $\Lambda = 2$. Average paired certification differences within each of 48 independent worlds. Their values lie in $[-1, 1]$, so the one-sided 95% Hoeffding lower bound is the mean minus $\sqrt{2\log(20)/48}$. The registered target requires this endpoint > 0.05. Other settings and the stratified t interval are descriptive; the standalone union is a comparator, not a jointly calibrated decision rule.

B. Whole-family queries on fixed views

The retrospective family study reuses the 48 main worlds and the stored threshold for .001. Its 24 comparison cells are three source constructions (OBS, projected R, projected Fusion) $\times$ two R sample budgets $\times$ two radii $\times$ finite/population information. Each cell retains 48 worlds $\times$ five repetitions, giving 5,760 comparison positions, not independent datasets. The underlying 2,880 finite and 288 population program records reduce to 2,160 distinct exact $(\mathcal{C}, w, \pi_0)$ requests. Both solvers run each request: 4,320 synthetic calls. Deduplication saves computation without dropping comparison positions.

We decode saved inputs exactly, normalize policy probabilities rationally and recheck each position's three saved duals. Among 240 primary finite-Fusion cases, 12 already have checked policy gains; 121 have checked witnesses constant across actions within each context, with $F(m) = 0$, constructed before complete family optimization. The remaining 107 are the fixed recovery set. This family-level uncertainty differs from a fixed policy lacking a positive lower bound. Theorem 1 permits selection on the inference view; this retrospective analysis is not a new confirmatory or out-of-sample learning test.

C. A complete exact solver and the source of recovery

SoPlex 8.1.0 in rational mode [24], [25], [26] solves the epigraph LP on identical $(\mathcal{C}, w, \pi_0)$:

$$\min_{m \in \mathcal{C}, t} \sum_x w_x \big(t_x - \pi_{0,x}^\top m_x\big), \qquad m_{xa} \leq t_x, \quad t_x \text{ free.} \tag{16}$$

Stationarity of free t_x gives $\sum_a \mu_{xa} = w_x$ for its nonnegative epigraph multipliers. Thus $\pi_{xa} = \mu_{xa}/w_x$ is a policy when $w_x > 0$; use the reference row when $w_x = 0$. Original-row duals and a feasible point complete the partition method's certificate contract. Import/export and checking are exact; neither the construction's answer nor a saved point is supplied.

Two lowest hashes per size cell select 78 unique requests (72 cells, 144 memberships) for three paired rounds in alternating order. Service runs from post-initialization transmission through checked certificates, including construction, verification, serialization and transport; startup, diagnostics and batch totals are separate. Fresh models/workers use one CPU, ten-second service and 8 GiB address-space limits; all phase limits are archived. Failures retain costs and denominators; ratios require complete pairs. Rounds measure machine variation, not independent samples.

Exact optima L_i^* for three normalized contrasts separate certificate transfer from policy choice: 7,762 unique $(\mathcal{C}, c)$ requests and 17,280 slots. Set $L_3^+ = \max(0, L_1^*, L_2^*, L_3^*)$, with zero from the reference. Here L_{saved} is the maximum of zero and saved lower bounds for these same three policies. Then

$$\mathcal{R} - L_{\text{saved}} = \underbrace{L_3^+ - L_{\text{saved}}}_{\text{certificate-transfer gap}} + \underbrace{\mathcal{R} - L_3^+}_{\text{three-policy restriction}} . \quad (17)$$

Both terms are nonnegative. Saved duals solve their stored objectives, but exact policy normalization can loosen transferred bounds. Missing optima leave the decomposition unknown. The 107-case recovery set and threshold remain unchanged.

Natural metadata partitions. Coat gender, jacket type and color each partition 300 items. Separate row families preserve the original 31-row views' exact coefficients, endpoints and confidence allocation. The evaluation includes 144 natural-row subviews, twelve box-only OBS positions, the 60-view grid and one full-cell control: 217 memberships, 125 unique requests. The construction supports 95; the generic LP covers 30 overlapping-row requests too. Subviews relax their parents and cannot increase capacity. Coverage uses each parent's event, without a joint 95% guarantee over all memberships. A feasible common point may exclude every policy; this is not exclusively a family-optimization benefit.

D. Repeated fixed-policy queries

The 63 inputs comprise three primary 1,000-query streams on one connected 512-cell view; 54 geometry streams ($d = 64, 512, 2048$; disjoint/mixed/connected rows); and six generated-log streams (64 actions, one million O and four million R events, uniform/skewed assignment). Broader streams have 100 queries each. Nearby queries perturb contrasts; unrelated directions preserve legal policies; $\delta = .001$ throughout. Primary streams have one solve and 999 reuse hits, all true; broader streams permit all three verdicts. These constructed streams do not estimate online hit rates; source ingestion is excluded.

Four plans cross screening S (deferring exact candidate checks) and retention P (saving exact dual constants/residuals across objectives): $P = 0$ discards precisely the terms retained by $P = 1$. All share native configuration, zero-dual/last-dual/primal order, proof format and receiver, with matching decisions, evidence and native paths. Optimized plans retain view decoding, primal state and within-request objective/box conversion; sparse residuals skip zeros. The separate last-first bridge uses $S_0 P_1$ arithmetic but checks last before zero, preserving decisions and native opportunities; true certificates may differ.

Joint service includes fresh producer/receiver initialization, view validation, queries including failures/native fallback, serialization and persistent FIFO16 verification; it excludes disk I/O, compression, network transfer and separate legacy/Fraction audits.

One quality pass (45,000 responses) and five timing rounds (225,000) use fresh arm-streams, one CPU/native thread and input-independent shuffled order. Enabled/disabled cost ratios fix the other switch; equal-input geometric means per group/round give four conditional ratios. Their log interaction and four bridge ratios are retained. Input ratios, slow cases, absolute costs, receiver shares and prefixes are archived. Registered descriptive checks ask whether retention helps without screening, and screening helps without retention but costs with it. Rounds measure machine variation, not independent workloads.

VI. Experimental Results

A. What complete family optimization adds

Broader policy search changes the fixed-view answer. The 240 primary finite Fusion views cover 48 worlds with five inference samples each. Saved evidence settles 12 with above-threshold policies and 121 with zero capacity. Full search resolves the remaining 107 views: 86 admit new above-threshold policies; 21 have $0 < \mathcal{R} \le \delta$, excluding every such guarantee. Positivity alone therefore need not justify further search. Overall, 98 views admit above-threshold policies and 142 do not. Exact reoptimization of the original three policies recovers none of the 86: their best value, including the reference, is zero in every case. Recovered guarantees have median 0.991 pp, interquartile range $[0.583, 1.549]$ pp and range $[0.105, 3.632]$ pp. Only policy choice changes; information and views remain fixed.

Both methods agree on 2,160 unique synthetic requests (5,760 comparison positions). Independent checking passes all 12,770/12,770 responses: 4,320 family responses, 7,762 fixed-policy optima, 220 Coat calls and 468 timing calls, without timeout or incomplete call (Fig. 2).

Exact answers with dimension-dependent cost. All 234 timing pairs succeed. Across three rounds, construction/SoPlex geometric-mean service ratios are 0.808–0.835 at $d = 8$, 1.066–1.129 at $d = 64$, and 1.170–1.196 at $d = 256$. The construction is slower in 5/78, 55/78 and 54/78 pairs, respectively. These compare checked Python rational construction with compiled rational LP. Pair costs, certificate sizes, memory observations and failure denominators are archived. Table II gives absolute costs. At $d = 256$, verification alone takes median 144.3/144.6 ms (Part/LP); certificates occupy about 149.5 kB.

Source information is a separate comparison. At the larger R budget with conservative radius, projected R passes 223/240 thresholds versus Fusion's 218/240; R has greater capacity in 128 pairs, Fusion in 100, with 12 ties. At the smaller R budget with oracle-boundary radius, OBS and

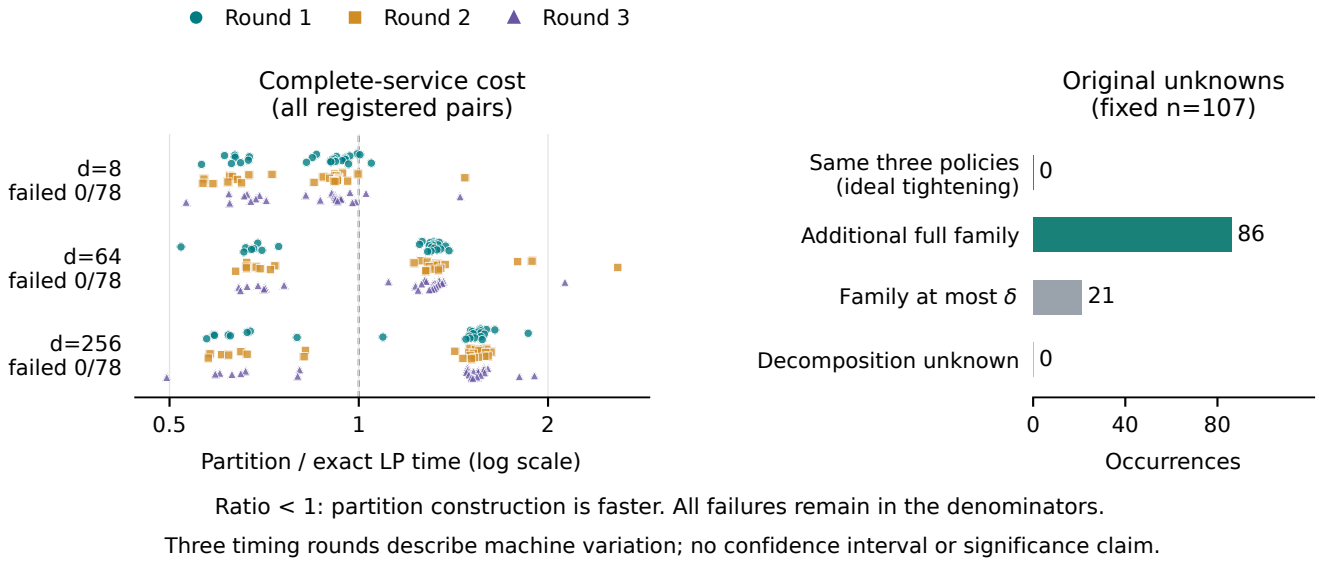


Fig. 2. Checked cost and fixed-view search. Left: 234 timing pairs (78 inputs, three rounds), from worker readiness through verification; startup/cleanup are separate. Both methods use the same receiver; offsets separate points. Right: 107 remaining primary Fusion views out of 240. *Same three policies*: exact reoptimization; *Additional full family*: a different policy guarantees gain $> \delta$; *Family at most* δ: no such guarantee exists. No failures or missing decompositions; rounds measure machine variation, not confidence intervals.

TABLE II
SEPARATE MEDIANS OVER 78 CALLS PER METHOD/DIMENSION, NOT PAIRED COST RATIOS. RESPONSE: AFTER WORKER INITIALIZATION THROUGH PROOF CHECKING; CHECK: PER-CALL RECEIVER SHARE; WHOLE: INCLUDES STARTUP/CLEANUP. PART: PARTITION CONSTRUCTION; LP: SOPLEX. KB: 1,000 BYTES.

d	Method	Response ms	Check %	Whole ms	kB
8	Part	4.66	24.4	122.2	3.47
8	LP	4.99	22.1	135.2	3.47
64	Part	53.58	23.2	172.2	20.93
64	LP	40.85	31.6	170.0	20.91
256	Part	648.13	22.8	770.3	149.51
256	LP	419.97	34.6	554.0	149.44

TABLE III
WHY A FIXED POLICY LACKS A GUARANTEE: 240 LOWER-BUDGET QUERIES PER POLICY AT $\Lambda = 2$. SAMPLING: EXACT POPULATION STATISTICS WOULD SUFFICE. AMBIGUITY: EVEN THOSE STATISTICS DO NOT SUFFICE, ALTHOUGH THE TRUE GAIN EXCEEDS δ. NO CASE IS MISSED SOLELY BY OPTIMIZATION.

Policy	Cert.	Sampling	Ambiguity	True $\leq \delta$
Metadata	12	33	195	0
R heuristic	4	43	190	3
Robust design	2	0	0	238

Fusion both pass 239 thresholds, but OBS has greater capacity in 219 pairs and Fusion in 21. Full- and split-budget views are not nested. All 24 comparison cells and exact decompositions remain available. Generating means satisfy all 2,880 finite partition views, and every returned policy assertion agrees with stored truth. These retrospective checks neither establish uniform coverage nor validate source assumptions.

B. The cost of answering repeated policy queries

All 270,000 responses pass independent exact replay; core plans match evidence, decisions and native paths. Across the five primary rounds (Fig. 3), retention ratios are 0.735–0.748 without screening and 0.766–0.791 with it. Screening ratios are 1.103–1.138 without retention and 1.165–1.186 with it. Screening also costs more in every geometry and source-derived group-round mean. Its earlier benefit without retention disappears under optimized arithmetic; two of three registered directions hold in every primary round.

The last-first bridge costs 0.833–0.882 times S_0P_1 on primary streams, but is slower in 1/15 primary, 66/270 geometry and 17/30 source-derived input-round pairs. Retention also has adverse individual cases despite lower group means. These results support retained terms on group averages, not default screening or universal speedups. The bridge separately changes candidate order.

a) Auxiliary execution controls: Under fixed arithmetic, screening plus retention fails the registered joint-cost-ratio target ≤ 0.8 against each feature alone. Screening slows every broader-group round. A larger bank misses its speed target; changed decisions preclude isolated reuse attribution. Materialization helps ingestion, but preparation costs 1.580/1.619 times hot execution under uniform/skewed assignment, respectively. Full controls remain in the artifact.

C. Source information and query guarantees

Here the policy is fixed and the source view changes, unlike the family search in VI-A. All 60 worlds and 1,200 inference configurations complete. For the metadata policy at $100d$ randomized events and $\Lambda = 2$, fusion certifies 12/240

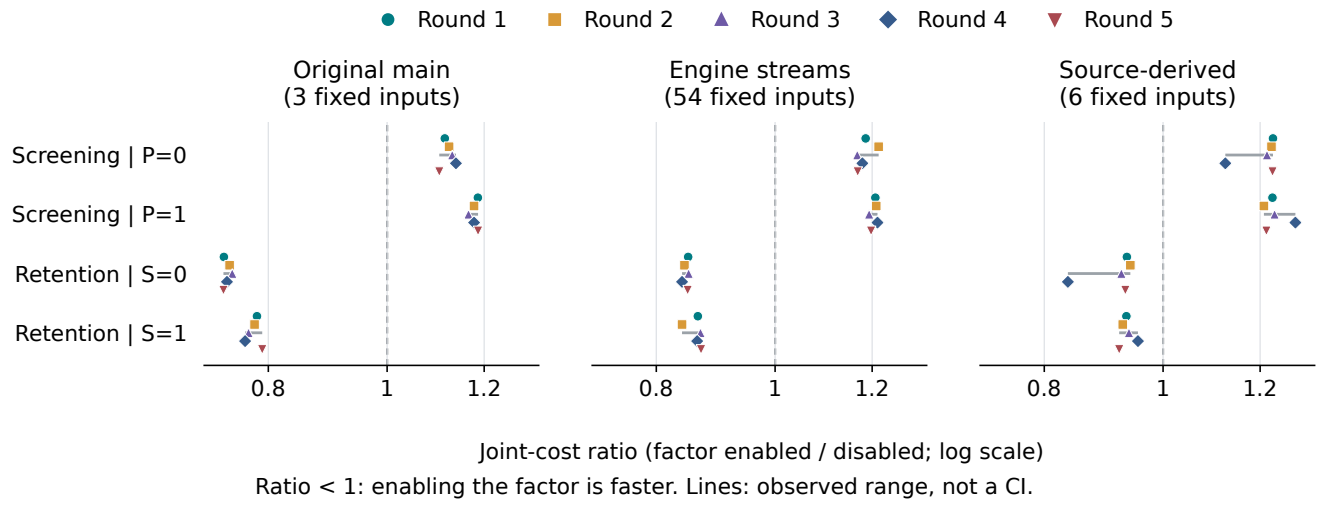


Fig. 3. Retention lowers group-average cost; screening does not. S: screening; P: retention (0 off, 1 on). Each row varies its named switch, fixing the other. Groups: *Original main*, primary; *Engine streams*, geometry; *Source-derived*, generated logs. Dots: each round's equal-input geometric mean of enabled/disabled cost ratios (below one is faster). Lines: five-round range, not confidence intervals. All three groups and four conditional comparisons appear; interaction and last-first comparisons are archived.

versus six for the standalone union (OBS six, R zero): six additions, no losses, certification-rate difference 0.025 (2.5 pp). Hoeffding's lower endpoint -0.3283 misses the registered target; its finite-world penalty limits power for modest gains. The descriptive t interval is $[-0.0249, 0.0749]$. All additions occur in two zero-drift 2×4 worlds. This does not confirm the target or establish absence of benefit.

Larger-budget projected fusion certifies 15 metadata queries, 16 heuristic queries and two robust-design selections. More randomized traffic does not resolve transfer or radius uncertainty.

Why adding a source need not improve the bound. For the same policy contrast c, fusion adds group constraints but splits the confidence budget. Let $\mathcal{B}_\alpha \subseteq \mathcal{B}_{\alpha/2}$ be full-budget OBS and fusion's observational box, with objective minima $B_\gamma = \min_{m \in \mathcal{B}_\gamma} c^\top m$. For nonempty $\mathcal{C}_F = \mathcal{B}_{\alpha/2} \cap \{a \le Tm \le b\}$, its optimum obeys

$$V_F - B_\alpha = \underbrace{(V_F - B_{\alpha/2})}_{G \ge 0} - \underbrace{(B_\alpha - B_{\alpha/2})}_{P \ge 0}. \tag{18}$$

Fusion improves the full-budget observational bound exactly when its constraint gain G exceeds its budget cost P.

Proposition 2 (A checkable zero-gain condition): Let $F_c = \arg\min_{m \in \mathcal{B}_{\alpha/2}} c^\top m$. Then $G = 0$ if and only if F_c contains a point satisfying all randomized group constraints. If F_c is a singleton, checking its moment rows decides whether $G = 0$ or $G > 0$.

Proof. A feasible box minimizer attains the box bound in $\mathcal{C}_F$. Conversely, compactness gives an optimizer in $\mathcal{C}_F$; equality places it in F_c. If the sets do not intersect, the gap is strict.□

Choose box endpoints by the signs of c and check the group rows. Feasibility proves zero constraint gain. If zero coefficients leave multiple minimizers, one failed point is inconclusive. We apply this test below.

Sampling uncertainty versus ambiguity. Finite views include sampling intervals; population views use exact generating statistics, retaining ambiguity. Among 240 primary queries with true effects above δ, ambiguity blocks 195 and sampling width 33; exact optima exclude optimization failure (Table III). Six additional finite certifications comprise four with sufficient standalone population information and two requiring fusion. Robust design retains the reference in 238/240 lower-budget cases: true effect zero.

Oracle radii use simulator truth. Across four radius/budget settings, fusion adds/loses 6/0, 9/0, 0/2 and 3/1 certifications versus the standalone union. Low-budget oracle metadata achieved gain averages 0.0114 pp versus 0.0913 pp budget cost; their difference changes the bound, not the policy effect. Proposition 2 proves zero optimal gain for 193/240 queries, including both losses; the larger-budget loss has positive gain below cost. All three losses persist at exact optima. Budget streams differ and robust policies can change across radii; fixed-world diagnoses estimate neither sample-size effects nor complementarity frequencies. Full diagnostics remain in the artifact.

a) Standalone-source controls: Observational consistency (5) lets OBS certify 99/100 external 2×2 queries; other policy/view settings differ. KuaiRec's imposed exposures and query-directed randomized means [27] do not establish general complementarity: a 35-setting census finds no true-radius fusion-only case. 400 count/policy pairs remain in the artifact.

The broader audit includes 6,000 inference views: 60 worlds, five repetitions, two R sample budgets, two radii and five constructions. The family's 2,880-view subset uses 48 main worlds and only OBS/projected-R/projected-Fusion. All 14 set misses occur outside that subset; none yields a bound above its true contrast or a false certification. All 18,900

final-mapping proofs (18,000 synthetic, 900 thinned Coat) pass independent checking; seven group-score certifications are unresolved after mapping, while projected statuses are unchanged. Separate design views have one miss in 600. Empirical coverage checks do not prove Theorem 1's assumptions.

D. Real data: policy redesign and stopping

Coat tests item selection for one population ($K = 1$), without personalization. Its 290 users each supply 24 self-selected and 16 randomized ratings on 300 items [28]. Success means rating at least four, not purchase; zero rating entries are unobserved. Users, not the 11,600 ratings, are independent. Potential ratings are assumed invariant to within-source order and panel composition; cross-source equality requires another assumption.

Gender, jacket-type and color partitions total 31 rows and overlap when combined. Twelve declarations use $\Lambda \in \{1, 1.5, 2, 3\}$ and $\rho \in \{0, .02, .05\}$; primary is $(2, 0)$. Fifteen candidates $\pi_i \propto t^{h_i}$, $t \in \{.5, 1.5, 2\}$ use five binary indicators: front-page placement, men's items, fleece, waterproof and black. The reference is uniform $1/300$. The primary policy doubles front-page weights and renormalizes. Only the three front-page contrasts lie outside the moment row space.

Keeping all ratings assumes each user's 16 R items form a simple random subset independent of potential ratings. The release says random/uniform; collection code is unavailable for verification. A value-independent control retains one rating per source/user: $q_a = (16/300)/16 = 1/300$. Its 580 events leave 116 empty O and 124 empty R cells; 900 queries and 15 full-cell controls yield no positive certificate. This diagnoses the cost of discarding panel structure.

For O, conditional CP uses included users per item; joint CP uses $N = 290$. Uniform thinning divides marginal and latent inclusion propensities by 24, preserving their ratio and conditional ratings. Full inclusion strengthens consistency without changing Λ, beyond reducing variance. Each R user with ratings in a group averages them. Under the subset design, this estimates their uniform group mean without bias; activity is independent of potential ratings. Conditional on active count, the iid scores lie in $[0, 1]$. Equation (8) uses 72–290 users; score variance uses denominator $n_j - 1$ and cannot be recovered from item histograms. Projected intervals use per-item CP; exact propagation follows (9), adding neither users nor transport evidence.

The fifteen candidates on 61 views give $15 \times 61 = 915$ queries. All lack positive certificates (`unresolved`); no view is refused. At the primary $(\Lambda, \rho) = (2, 0)$ declaration, group Fusion improves full-budget R group's bound by 0.309 pp, but neither passes $\delta = .001$ (Table IV). Independent audit reconstructs user statistics, 61 views, 915 proofs and 15 direct controls; it does not establish source assumptions.

Stopping under the primary declaration. All five primary $\Lambda = 2, \rho = 0$ views have zero capacity. Group Fusion permits all item means to equal 0.2186 (rounded), tying every policy with the reference. This checked constant point justifies stopping without full optimization. The stronger full-cell control also has zero capacity.

TABLE IV
COAT: PRIMARY FIXED-QUERY LOWER BOUNDS (PROBABILITY UNITS; $\Lambda = 2$, $\rho = 0$, $\alpha = .05$; 290 USERS) AND FAMILY COUNTS SUMMING TO TWELVE DECLARATIONS. POSITIVE FAMILY ANSWERS OCCUR OUTSIDE THE PRIMARY DECLARATION. FULL-CELL TRANSPORT IS A STRONGER CONTROL.

	One policy	All policies over declarations		
Construction	Primary LCB	$\mathcal{R} = 0$	$0 < \mathcal{R} \le \delta$	$\mathcal{R} > \delta$
OBS	-0.03729	9/12	0/12	3/12
R group	-0.01502	12/12	0/12	0/12
R projected	-0.03402	12/12	0/12	0/12
Fusion group	-0.01193	7/12	1/12	4/12
Fusion projected	-0.03132	9/12	3/12	0/12
R full-cell control	-0.03077	1/1	0/1	0/1

Searching beyond candidates under other declarations. With ideal uniform reference $1/300$, exact family optimization resolves 60 grid positions: 49 have $\mathcal{R} = 0$, four $0 < \mathcal{R} \le .001$, and seven $\mathcal{R} > .001$. The seven positive positions are five unique requests: OBS at $\Lambda = 1$ repeats across three tolerances; group Fusion passes at $\Lambda = 1$ for all three tolerances and at $\Lambda = 1.5, \rho = 0$. The original fifteen candidates certified none. Guarantees remain conditional on the stated assumptions, without joint coverage across declarations.

A redesign under stronger assumptions ($\Lambda = 1.5, \rho = 0$). The jacket-type certificate moves all 28 vests' uniform mass to item 0, a men's bomber: its probability rises from $1/300$ to $29/300$; vests receive zero and 271 other items retain $1/300$. All 15 original and 50 extended catalogue policies retain full support; none makes this choice. The contrast is $(28/300)(m_0 - \overline{m}_{\text{vest}})$, where $\overline{m}_{\text{vest}}$ is the 28 vests' uniform mean. Its checked lower bound is $(28/300)(\ell_0 - b_{\text{vest}}/\kappa_{\text{vest}})$: row mass κ converts the group constraint to its uniform mean. The only nonzero terms, 2.472638 pp and -2.226283 pp, sum to 0.246355 pp in expected high-rating probability over uniform. This exceeds 0.1 pp and matches the full overlapping view's capacity; matched full-budget OBS and both R views have zero capacity. This retrospective guarantee assumes confounding no greater than $\Lambda = 1.5$ and no group-mean drift. It neither selects assumptions from data nor measures purchase uplift, and does not imply every vest is worse.

Separate gender, jacket-type and color partitions give 144 view memberships: 10 exceed the threshold, 10 have positive capacity at most the threshold, and 124 have zero capacity. Construction and exact LP agree on 95 unique supported requests across the full evaluation; LP also resolves 30 overlaps. Subview capacity cannot exceed its parent's, as relaxation requires. These are dependent views of one dataset, not 144 independent experiments or new evidence for sampling assumptions.

Under full-cell transport, a separate direct control averages all 16 R ratings within each user and applies a one-sided Hoeffding bound ($N = 290$). The primary estimate is 0.00371,

TABLE V
PREPROCESSING THE UNCHANGED COAT CATALOGUE. TIME: MEDIAN SECONDS OVER THREE ROUNDS, INCLUDING STARTUP, FAMILY WORK, EXTERNAL VERIFICATION AND CLEANUP. RATIO/OPT: PAIRED COST RANGE (ABOVE ONE IS SLOWER). N/E/F: NATIVE/FIXED-POLICY EXACT/FAMILY CALLS, IDENTICAL ACROSS QUALITY AND TIMING PER STREAM. BOTH VIEWS USE $\rho = 0$.

View	Plan	Time (s)	Ratio/Opt	N/E/F
$\Lambda = 1.5$	Hot	6.800	0.974–1.009	2/2/0
$\Lambda = 1.5$	Opt	6.934	1.000	2/2/0
$\Lambda = 1.5$	Opt+Part	7.312	1.051–1.077	2/2/1
$\Lambda = 1.5$	Opt+LP	7.276	1.042–1.079	2/2/1
$\Lambda = 2$	Hot	6.329	0.985–0.998	1/1/0
$\Lambda = 2$	Opt	6.390	1.000	1/1/0
$\Lambda = 2$	Opt+Part	6.431	0.999–1.024	1/1/0
$\Lambda = 2$	Opt+LP	6.359	0.990–1.002	1/1/0

with lower bound -0.06452; none of 15 fixed queries passes. These per-query bounds neither support adaptive selection nor validate transport or a zero effect.

a) Catalogue queries with ordinary caching: At primary $(\Lambda, \rho) = (2, 0)$, each of five source views receives an extended catalogue combining two indicators: $\pi_i \propto t_1^{h_{1i}} t_2^{h_{2i}}$, $(t_1, t_2) \in \{.5, 1, 2\}^2 \setminus \{(1, 1)\}$. Ten pairs give 80 policy positions at margins $0, .001, .01$ (0, 0.1, 1 pp): 240 requests per view, 50 objectives and 150 objective–margin pairs. Prepared execution retains exact dual terms; optimized-exact adds sparse arithmetic and cached number conversions. Both fully cache objectives and retain a persistent receiver. Native hot-start retains a solver basis without cross-objective certificate candidates. Quality checks precede three timing rounds.

Prepared and optimized-exact agree on all 1,200 requests. Optimized/prepared cost ratios 0.960, 0.941, 0.953 miss the ten-percent target; optimized-exact is slower in 3/15 view-rounds and retains more estimated exact-state payload in every view (not isolated RSS). Native hot-start changes 98 false answers to unknown and four unknown answers to false, never reversing a Boolean answer. Overall service differs; on R projected, decisions agree and native is faster in all rounds. None certifies improvement; neither fusion-only information target is supported. All five primary views have zero capacity, already checkable by a constant-action witness.

b) The cost of family preprocessing: A redesign outside the catalogue need not certify any catalogue policy. We test whether family evidence instead reduces cost for the unchanged 240 requests, using two 16-row jacket-type Fusion views, $(\Lambda, \rho) = (1.5, 0)$ and $(2, 0)$, selected retrospectively with known capacities. Hot-start (Hot), optimized threshold execution (Opt), and Opt with partition (Part) or exact-LP (LP) preprocessing share objective caching, checked points and a rational receiver. Family certificates add only feasible points; a shared constant-action check skips unnecessary family solves. Unresolved stored-contrast queries allow one producer attempt per objective–margin pair and one complete exact fallback per objective; ideal rational policies reuse that evidence without extra solves.

All 32 streams complete with 240 false answers each, for both stored numeric contrasts and ideal rational policies. At nonzero capacity, Part adds 5.1–7.7% to Opt's full cost without reducing its two native and two exact calls (Table V); LP saves no calls either. At zero capacity, family computation is skipped. Three rounds measure machine variation; execution positions are not fully balanced. Optimal points differ, so costs compare complete preprocessing choices. Family answers enable redesign beyond the catalogue but do not amortize preprocessing here.

VII. RELATED WORK AND SCOPE

Database queries. Causal systems adjust aggregates [1], implement matching [2], define units [29] and explain patterns [30]. Other approaches propagate uncertainty bounds [3] or certify semantic-view predicates from fresh draws [31]. Our source-constrained view supports adaptive objectives under one coverage event and whole-family stopping certificates.

Our view combines binary density-ratio and conditional-moment bounds [5], [6], without claiming that every allowed vector can arise from a full latent-variable model. Other fusion methods exploit overlap [32], impose experimental moments [33], or estimate sensitivity frontiers [34]. Propensity-odds policy learning [35], [36] makes different restrictions from our outcome-conditional weights. We add direct partition construction and original-constraint certificate recovery in one query interface.

Classical allocation. Continuous maximin sharing [37, Sec. 2.2] solves $\max_{0 \le z \le 1} \min_g \sum_{i \in G_g} v_i z_i$ under one resource budget, recovering capacities and allocations. Its groups partition profits; ours impose disjoint two-sided moments on means. We minimize $\max_i m_i - p^\top m$ per context and recover original-constraint certificates. The general exact LP supplies the same family answer and certificate contract.

Proofs and scope. Filtering [38], assignment caching [39], safe bounds and reuse are established; VIPR checks exact optimization derivations [13]. Unlike IntegriDB [40], our digests authenticate neither data nor freshness. Scope is binary outcomes, known target weights and trusted source/interval routines; continuous covariates, estimated weights and dependent interactions need extensions.

Data ethics. We reanalyze public benchmarks under their release terms, without recruiting or contacting participants. No institutional ethics review was undertaken or additional consent sought.

VIII. CONCLUSION

An uncertified policy need not end policy search. Q-BRIDGE supports optimal redesign and stopping through a shared confidence view and independently checked certificates. Exact-term retention lowers group-average cost on the tested streams.

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OpenAI Codex assisted with drafting and mathematical exposition (Sections I–VIII), code and experiments (III–VI), figures, source retrieval and editing. Computational claims were checked against saved artifacts and citations against primary sources. The authors remain responsible for correctness, originality and the final submission.

References

[1] B. Salimi, J. Gehrke, and D. Suciu, "Bias in OLAP queries: Detection, explanation, and removal," in *Proceedings of the 2018 International Conference on Management of Data*. ACM, 2018, pp. 1021–1035. [Online]. Available: https://doi.org/10.1145/3183713.3196914

[2] B. Salimi, C. Cole, D. R. K. Ports, and D. Suciu, "ZaliQL: Causal inference from observational data at scale," *Proceedings of the VLDB Endowment*, vol. 10, no. 12, pp. 1957–1960, 2017. [Online]. Available: https://www.vldb.org/pvldb/vol10/p1957-salimi.pdf

[3] S. Feng, A. Huber, B. Glavic, and O. Kennedy, "Uncertainty annotated databases—a lightweight approach for approximating certain answers," in *Proceedings of the 2019 International Conference on Management of Data*. Association for Computing Machinery, 2019, pp. 1313–1330. [Online]. Available: https://www.cs.uic.edu/~bglavic/dbgroup/assets/pdfpubls/FH19.pdf

[4] J. Fawkes, M. O'Riordan, A. Vlontzos, O. Corcoll, and C. M. Gilligan-Lee, "The hardness of validating observational studies with experimental data," in *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 258. PMLR, 2025, pp. 1819–1827. [Online]. Available: https://proceedings.mlr.press/v258/fawkes25b.html

[5] J. Dorn, K. Guo, and N. Kallus, "Doubly-valid/doubly-sharp sensitivity analysis for causal inference with unmeasured confounding," *Journal of the American Statistical Association*, vol. 120, no. 549, pp. 331–342, 2025. [Online]. Available: https://doi.org/10.1080/01621459.2024.2335588

[6] K. Ishikawa, N. He, and T. Kanamori, "A convex framework for confounding robust inference," *Journal of Machine Learning Research*, vol. 27, no. 62, pp. 1–53, 2026. [Online]. Available: https://jmlr.org/papers/v27/23-1434.html

[7] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934. [Online]. Available: https://doi.org/10.1093/biomet/26.4.404

[8] A. Maurer and M. Pontil, "Empirical Bernstein bounds and sample variance penalization," in *Proceedings of the 22nd Annual Conference on Learning Theory*, Montreal, Canada, 2009. [Online]. Available: https://www.cs.mcgill.ca/~colt2009/papers/012.pdf

[9] R. Berk, L. Brown, A. Buja, K. Zhang, and L. Zhao, "Valid post-selection inference," *The Annals of Statistics*, vol. 41, no. 2, pp. 802–837, 2013. [Online]. Available: https://arxiv.org/abs/1306.1059

[10] D. Bertsimas, V. Gupta, and N. Kallus, "Data-driven robust optimization," *Mathematical Programming*, vol. 167, no. 2, pp. 235–292, 2018.

[11] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge University Press, 2004. [Online]. Available: https://web.stanford.edu/~boyd/cvxbook/

[12] A. Neumaier and O. Shcherbina, "Safe bounds in linear and mixed-integer linear programming," *Mathematical Programming*, vol. 99, no. 2, pp. 283–296, 2004. [Online]. Available: https://arnold-neumaier.at/ms/mip.pdf

[13] K. K. H. Cheung, A. Gleixner, and D. E. Steffy, "Verifying integer programming results," in *Integer Programming and Combinatorial Optimization*, ser. Lecture Notes in Computer Science, vol. 10328. Springer, 2017, pp. 148–160. [Online]. Available: https://arxiv.org/abs/1611.08832

[14] M. Sion, "On general minimax theorems," *Pacific Journal of Mathematics*, vol. 8, no. 1, pp. 171–176, 1958. [Online]. Available: https://msp.org/pjm/1958/8-1/pjm-v8-n1-p14-s.pdf

[15] H. A. Eiselt, "Continuous maximin knapsack problems with GLB constraints," *Mathematical Programming*, vol. 36, pp. 114–121, 1986.

[16] T. Kuno, H. Konno, and E. Zemel, "A linear-time algorithm for solving continuous maximin knapsack problems," *Operations Research Letters*, vol. 10, no. 1, pp. 23–26, 1991.

[17] T. Marcucci and R. Tedrake, "Warm start of mixed-integer programs for model predictive control of hybrid systems," *IEEE Transactions on Automatic Control*, vol. 66, no. 6, pp. 2433–2448, 2021. [Online]. Available: https://groups.csail.mit.edu/robotics-center/public_papers/Marcucci19.pdf

[18] H. Kothari and J. R. Luedtke, "Accelerating Benders decomposition for solving a sequence of sample average approximation replications," *Mathematical Programming Computation*, 2026, published online May 23, 2026. [Online]. Available: https://link.springer.com/article/10.1007/s12532-026-00324-3

[19] A. Deza, F. Badilla, and S. S. Dey, "Mixed-integer program sensitivity analysis via branch-and-bound trees," 2026, preprint, version 1, submitted November 3, 2026. [Online]. Available: https://arxiv.org/abs/2611.04415v1

[20] HiGHS developers, "HiGHS documentation: Hot start and python interfaces," Official software documentation, 2026, accessed 2026-11-04; API cross-checked with HiGHS v1.12.0 source and local HiGHS 1.12.0 and 1.15.1 bindings. [Online]. Available: https://ergo-code.github.io/HiGHS/dev/guide/further/

[21] ——, "HiGHS documentation: List of options," 2026, official documentation, accessed 2026-11-04; option semantics checked against local HiGHS 1.15.1. [Online]. Available: https://ergo-code.github.io/HiGHS/dev/options/definitions/

[22] S. Martello and P. Toth, *Knapsack Problems: Algorithms and Computer Implementations*, ser. Wiley-Interscience Series in Discrete Mathematics and Optimization. Chichester, UK: John Wiley & Sons, 1990. [Online]. Available: https://silvano333.github.io/kp.html

[23] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Mathematical Programming Computation*, vol. 10, no. 1, pp. 119–142, 2018. [Online]. Available: https://doi.org/10.1007/s12532-017-0130-5

[24] Zuse Institute Berlin, "SoPlex 8.1.0," https://soplex.zib.de/, 2026, released September 2026. Accessed November 6, 2026.

[25] A. M. Gleixner, D. E. Steffy, and K. Wolter, "Iterative refinement for linear programming," *INFORMS Journal on Computing*, vol. 28, no. 3, pp. 449–464, 2016.

[26] A. Gleixner and D. E. Steffy, "Linear programming using limited-precision oracles," *Mathematical Programming*, vol. 183, pp. 525–554, 2020.

[27] C. Gao, S. Li, W. Lei, J. Chen, B. Li, P. Jiang, X. He, J. Mao, and T.-S. Chua, "KuaiRec: A fully-observed dataset and insights for evaluating recommender systems," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. ACM, 2022, pp. 540–550. [Online]. Available: https://kuairec.com/

[28] T. Schnabel, A. Swaminathan, A. Singh, N. Chandak, and T. Joachims, "Recommendations as treatments: Debiasing learning and evaluation," in *Proceedings of the 33rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 48. PMLR, 2016, pp. 1670–1679. [Online]. Available: https://proceedings.mlr.press/v48/schnabel16.html

[29] B. Salimi, H. Parikh, M. Kayali, L. Getoor, S. Roy, and D. Suciu, "Causal relational learning," in *Proceedings of the 2020 ACM SIGMOD International Conference on Management of Data*. ACM, 2020, pp. 241–256. [Online]. Available: https://doi.org/10.1145/3318464.3389759

[30] B. Youngmann, M. Cafarella, A. Gilad, and S. Roy, "Summarized causal explanations for aggregate views," *Proceedings of the ACM on Management of Data*, vol. 2, no. 1, pp. 71:1–71:27, 2024. [Online]. Available: https://doi.org/10.1145/3639328

[31] O. Hassanzadeh and D. Subramanian, "Semantic views: Certified reuse of LLM labels across semantic queries," 2026, preprint, version 1, submitted October 8, 2026. [Online]. Available: https://arxiv.org/abs/2610.12768v1

[32] K. Jiang, W. Hu, X. Lai, S. Yang, and X.-H. Zhou, "Improving sensitivity analysis by synthesizing randomized clinical trials with limited overlap," *Statistics in Medicine*, vol. 45, no. 18–19, p. e70637, 2026. [Online]. Available: https://onlinelibrary.wiley.com/doi/10.1002/sim.70637

[33] Y. Du, Z. Zhang, H. Li, C. Fang, J. Xu, Z. Peng, and J. Guo, "Feasible fusion: Constrained joint estimation under structural non-overlap," in *Proceedings of the 43rd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 306. PMLR, 2026, pp. 26 851–26 885. [Online]. Available: https://proceedings.mlr.press/v306/du26s.html

[34] Q. Lanners, C. Rudin, A. Volfovsky, and H. Parikh, "Data fusion for partial identification of causal effects," in

Advances in Neural Information Processing Systems, vol. 38. Curran Associates, Inc., 2025, pp. 150 638–150 702. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/hash/dd4e79c2228c43954cc074abf2f85a07-Abstract-Conference.html

[35] N. Kallus and A. Zhou, "Confounding-robust policy improvement," in *Advances in Neural Information Processing Systems*, vol. 31, 2018. [Online]. Available: https://proceedings.neurips.cc/paper/2018/hash/3a09a524440d44d7f19870070a5ad42f-Abstract.html

[36] K. Hess, D. Frauen, V. Melnychuk, and S. Feuerriegel, "Efficient and sharp off-policy learning under unobserved confounding," in *International Conference on Learning Representations*, 2026, pp. 57 552–57 587. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2026/file/5e972fe492946fc7a02aef64d0f4fd7e-Paper-Conference.pdf

[37] T. Yamada, M. Futakawa, and S. Kataoka, "Some exact algorithms for the knapsack sharing problem," *European Journal of Operational Research*, vol. 106, no. 1, pp. 177–183, 1998. [Online]. Available: https://www.nda.ac.jp/~yamada/paper/98-2.pdf

[38] T. Bartels, V. Fisikopoulos, and M. Weiser, "Fast floating-point filters for robust predicates," *BIT Numerical Mathematics*, vol. 63, 2023, article 31. [Online]. Available: https://doi.org/10.1007/s10543-023-00975-x

[39] C. Cadar, D. Dunbar, and D. Engler, "KLEE: Unassisted and automatic generation of high-coverage tests for complex systems programs," in *USENIX Symposium on Operating Systems Design and Implementation*, 2008. [Online]. Available: https://static.usenix.org/event/osdi08/tech/full_papers/cadar/cadar_html/paper.html

[40] Y. Zhang, J. Katz, and C. Papamanthou, "IntegriDB: Verifiable SQL for outsourced databases," in *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security*. Association for Computing Machinery, 2015, pp. 1480–1491. [Online]. Available: https://integridb.github.io/IntegriDB.pdf